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COURSE NAME – Operation System
COURSE CODE – CSA0408

Lab Observations

EXPERIMENT - 10

Illustrate the concept of inter-process communication using message queue with a C program.

PROGRAM:

```
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <unistd.h>
#include <sys/types.h>
#include <sys/ipc.h>
#include <sys/msg.h>

struct message
{
    long msg_type;
    char msg_text[100];
};

int main()
{
    key_t key;
```

```
int msgid;
struct message msg;

/* Generate a unique key */
key = ftok("msgqfile", 65);

/* Create message queue */
msgid = msgget(key, IPC_CREAT | 0666);

if(msgid == -1)
{
    perror("msgget");
    exit(EXIT_FAILURE);
}

/* Set message type */
msg.msg_type = 1;

/* Producer: Send message */
strcpy(msg.msg_text, "Hello, message queue!");

if(msgsnd(msgid, &msg, sizeof(msg.msg_text), IPC_NOWAIT) == -1)
{
    perror("msgsnd");
    exit(EXIT_FAILURE);
}
```

```
    printf("Producer: Data sent to message queue: %s\n",
           msg.msg_text);

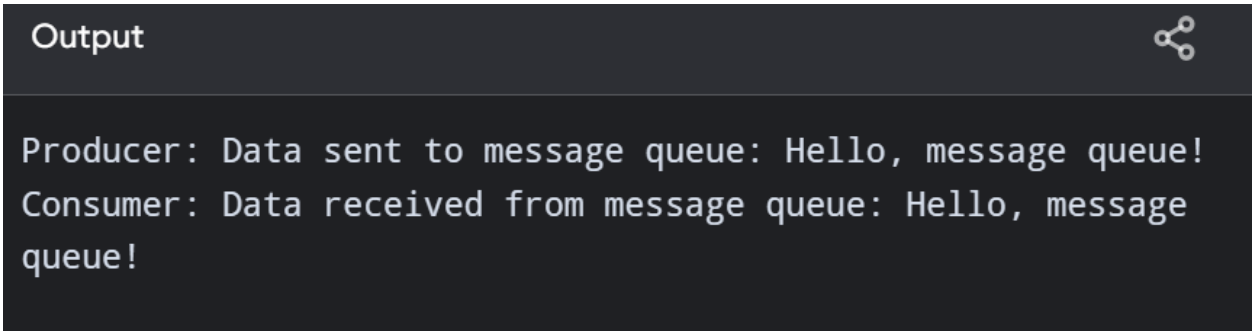
    /* Consumer: Receive message */
    if(msgrcv(msgid, &msg, sizeof(msg.msg_text), 1, 0) == -1)
    {
        perror("msgrcv");
        exit(EXIT_FAILURE);
    }

    printf("Consumer: Data received from message queue: %s\n",
           msg.msg_text);

    /* Remove message queue */
    if(msgctl(msgid, IPC_RMID, NULL) == -1)
    {
        perror("msgctl");
        exit(EXIT_FAILURE);
    }

    return 0;
}
```

OUTPUT:


Output
Producer: Data sent to message queue: Hello, message queue!
Consumer: Data received from message queue: Hello, message queue!